## Plasmids

Apart from the bacterial chromosome, bacteria also contain small circular, double stranded DNA molecules called plasmids (Figure 7.13). Plasmids are self replicating extra chromosomal genetic elements. Plasmids may carry genes for activities such as antibiotic resistance and tolerance to toxic metals. Examples: Fertility plasmid (F plasmid), Resistance plasmid (R plasmid) and colicin plasmid (Col plasmid).

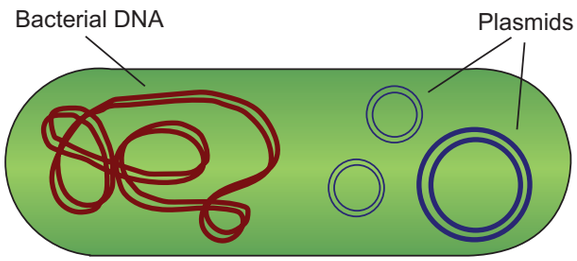



**Figure 7.13:** Plasmids in Prokaryotes

## Molecular Chaperones

They are the helper proteins which recognize the newly formed polypeptides and fold them into their proper shape of secondary and tertiary structure. Many chaperones are involved in proper folding of bacteria. They were first identified in *Escherichia coli* mutant. Example: Heat shock proteins are produced in *Escherichia coli* cells subjected to live at high temperatures, or in any other stressful unfavorable conditions.

## Inclusions

The cytoplasm of prokaryotic cells has several kinds of reserve deposits known as inclusions. Cells may accumulate certain nutrients when they are plentiful and use them when they are deficient. Some inclusions are common to a wide variety of bacteria whereas others are limited to certain species (Table 7.5).

## Endospores:

Some species of bacteria produce metabolically dormant structures called spores. They are highly durable and dehydrated resting bodies produced inside the cells. They are formed by bacteria only when there is lack of water or depletion of essential nutrients in the environment. Endospores are coated with a specific chemical compound diaminopimelic acid. It binds with the Calcium and forms Calcium dipicolinate which removes the water from it and makes the spore resistant to extreme conditions Example: *Bacillus anthracis* and *Clostridium tetani* possess endospores.

## Mesosomes

Generally prokaryotes do not have cytoplasmic organelles like mitochondria and chloroplast. It contains mesosome as their organelle. They are the invaginations of the cell membrane and they are in the form of tubules, vesicles or lamellae. They are seen in both Gram positive and Gram negative bacteria, generally more in Gram positive bacteria. They are located next to the septa or cross walls in dividing bacteria (Figure 7.14). They may be involved in cell wall formation during division or play a role in chromosome replication and distribution to daughter cells. If they are located near to the surface they are called peripheral mesosomes and if they are located deep into the cytoplasm they are called central mesosomes.

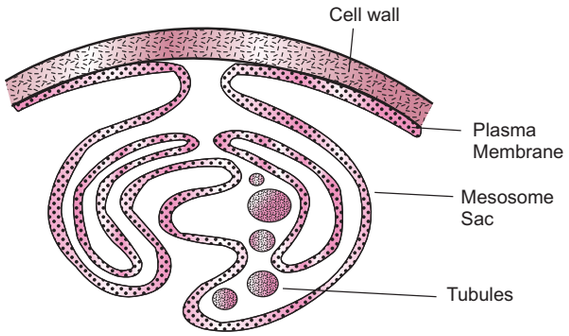



**Figure 7.14:** Bacterial Mesosome



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